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L06 Regression and Classification Models

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Reflection on Regression and Classification

In this lab, our group explored how regression and classification models use data to make different types of predictions. We worked with the Titanic dataset and used Linear Regression to predict passenger fares and Logistic Regression to predict whether passengers survived. Although both models followed similar steps, they had different goals and required different ways to evaluate their results. This activity helped us apply the data preparation techniques from our previous lab to actual machine learning models. We also learned that training a model is only one part of the process because understanding its predictions and evaluating its performance matter just as much.

Working with Linear Regression showed us how a model can use numerical features to estimate a continuous value (Tien, Zaid, and Viktoriya). We used Age and Pclass to predict passenger fares, trained the model, and made predictions using the test data (Tien and Zaid). One thing we noticed was that training a model only represents part of the process because we still need to check how well it performs (Viktoriya). When we evaluated our model, we received a Mean Squared Error of 3364.92 and a Root Mean Squared Error of 58.01 (Viktoriya). At first, we found MSE a little confusing because it squares the differences between actual and predicted fares (Viktoriya). Comparing it with RMSE helped us understand the results better because RMSE shows the error in the same units as the fare (Viktoriya). This helped us realize why choosing an understandable evaluation metric matters when explaining model performance (Viktoriya).

The model's coefficients gave us another way to understand what Linear Regression learned from the dataset (Zaid). We examined the coefficients for Age and Pclass and found that both had negative values. The coefficient for Age was approximately -0.44, while Pclass had a coefficient of approximately -38.30 (Zaid). These values showed us how the model's predicted fare changes when one feature increases while the other stays the same. However, we also learned that these coefficients describe relationships within the model rather than proving that one feature directly causes a change in fare (Zaid). This made us think more carefully about how we interpret model results instead of looking only at the final predictions (Zaid).

Logistic Regression introduced a different goal because we wanted to predict a category rather than a numerical value (Tien and Viktoriya). We used Age, Pclass, and Sex to predict passenger survival, and we converted Sex into numerical values so the model could use that feature (Tien and Viktoriya). After creating and training the model, we made predictions and

evaluated its accuracy (Viktoriya). Our model achieved 74.83% accuracy, which means it correctly classified roughly three out of four passengers in the test set (Viktoriya). This result helped us understand that a model can perform reasonably well without making every prediction correctly (Viktoriya). At the same time, accuracy alone did not tell us which types of passengers the model struggled to classify (Viktoriya). This raised questions about whether the model made more mistakes when predicting survivors or non-survivors (Viktoriya).

The Knowledge Check helped us connect both tasks and understand why we need different evaluation methods for each one (Tien, Zaid, and Viktoriya). Regression predicts continuous numerical values, so Mean Squared Error measures how far the predictions are from the actual values (Tien and Viktoriya). Classification predicts categories, so accuracy measures how often the model chooses the correct class (Tien and Viktoriya). We realized that accuracy would not work well for fare prediction because even a small difference between the actual and predicted fare would count as an incorrect result (Viktoriya). This also reminded us that no single evaluation metric works best for every machine learning problem (Viktoriya). We need to think about what we want to predict and what the results actually tell us about the model (Viktoriya).

The lab reminded us why we need to keep our notebook organized and check our work when several people handle different tasks (Tien, Zaid, and Viktoriya). We adjusted the Markdown formatting, reviewed the code, and checked the results to make sure the notebook remained clear and consistent (Tien). We also saw how each coding step depended on earlier steps, from preparing the data to training the models and evaluating their predictions (Tien, Zaid, and Viktoriya). Another interesting difference from our previous lab was how we handled missing Age values. In Lab 05, we used median imputation, but in this notebook, we removed rows that had missing Age values. This made us wonder whether keeping those rows through imputation would change the models' performance (Tien, Zaid, and Viktoriya). We would also like to test additional features and compare other evaluation metrics, such as precision and recall, to better understand the classification results (Viktoriya).

In conclusion, this lab helped our group move from preparing datasets to understanding how machine learning models use them to make predictions. Linear Regression taught us how to predict numerical values, while Logistic Regression showed us how to classify observations into categories. By evaluating both models, we learned that making predictions is not enough because we also need to check how well the models perform. Moreover, we learned that coefficients, evaluation metrics, and data preparation choices all affect how we interpret the results. Most importantly, this lab encouraged us to look beyond whether the code runs and think about what the model actually learned, how reliable its predictions are, and what we could improve next time.